

Daily Geometry Drill #4 — Answers & Investigations

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Exterior angles sum to 360° · Isosceles chains · Regular polygons as equal triangles · Similar triangles · Area ratios.

Section A Rapid Recognition

1. Write down the size of each interior angle of a regular 12-sided polygon.

Answer: 150°

Method: Each exterior angle is $\frac{360^\circ}{12} = 30^\circ$, so each interior angle is $180^\circ - 30^\circ = 150^\circ$.

Investigate further: Which regular polygons have an interior angle that is a whole number of degrees? (There are 22 of them; the exterior angle must divide 360° .)

2. An isosceles triangle has apex angle 40° . Write down each base angle.

Answer: 70°

Method: The base angles are equal and sum to $180^\circ - 40^\circ = 140^\circ$, so each is 70° .

Investigate further: An isosceles triangle has one angle of 40° , but you are not told which. List every possibility for the other two angles.

3. The incircle of the right triangle with legs 3 and 4 touches all three sides. What is the inradius?

Answer: 1

Method: The incircle is tangent to all three sides. For a right triangle with legs 3, 4 the hypotenuse is 5 and $r = \frac{a+b-c}{2} = \frac{3+4-5}{2} = 1$.

Investigate further: Show that $r = \frac{a+b-c}{2}$ for every right-angled triangle, by marking the equal tangent lengths from each vertex to the incircle.

4. Each interior angle of a regular polygon is 162° . Write down the number of sides.

Answer: 20

Method: Each exterior angle is $180^\circ - 162^\circ = 18^\circ$, and the exterior angles sum to 360° , so there are $\frac{360}{18} = 20$ sides.

Investigate further: Can a regular polygon have interior angle 155° ? Explain using the exterior angle.

5. Write down the number of diagonals of a decagon.

Answer: 35

Method: Each of the 10 vertices joins to $10 - 3 = 7$ others by a diagonal (not itself or its two neighbours), and each diagonal is counted twice: $\frac{10 \cdot 7}{2} = 35$.

Investigate further: Which polygon has exactly as many diagonals as sides?

6. The angle bisector from A of triangle ABC meets BC at D , with $AB : AC = 2 : 3$ and $BC = 10$. Write down DC .

Answer: 6

Method: The angle bisector theorem: $BD : DC = AB : AC = 2 : 3$ with $BD + DC = 10$, so $DC = \frac{3}{2+3} \cdot 10 = 6$.

Investigate further: Prove the angle bisector theorem by comparing the areas of triangles ABD and ACD in two ways.

7. The isosceles triangle with sides 5, 5, 6 has base 6. Write down the median from the apex to the base.

Answer: 4

Method: The median to the base is the altitude: $\sqrt{5^2 - 3^2} = \sqrt{16} = 4$.

Investigate further: Find the median from one of the base vertices of the same triangle. (Use Apollonius: $m^2 = \frac{2b^2 + 2c^2 - a^2}{4}$.)

8. A median of a triangle has length 12. How far is the centroid from the vertex at the end of that median?

Answer: 8

Method: The centroid is $\frac{2}{3}$ of the way from the vertex, so the vertex-to-centroid distance is $\frac{2}{3} \cdot 12 = 8$.

Investigate further: The three medians split a triangle into six small triangles. Show that all six have the same area.

9. Two similar triangles have corresponding sides in the ratio 2 : 3. The smaller has area 8. Write down the area of the larger.

Answer: 18

Method: Areas scale as the square of the length ratio: $8 \cdot \left(\frac{3}{2}\right)^2 = 18$.

Investigate further: Two similar solids have lengths in the ratio 2 : 3. What is the ratio of their volumes, and of their surface areas?

10. Write down the area of a regular hexagon with side 2.

Answer: $6\sqrt{3}$

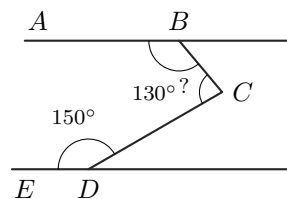
Method: A regular hexagon is six equilateral triangles of side 2, each of area $\frac{\sqrt{3}}{4} \cdot 4 = \sqrt{3}$. Total $6\sqrt{3}$.

Investigate further: Show that a regular hexagon of side s has area exactly $\frac{3}{2}$ times that of the equilateral triangle on its long diagonal.

Section B Manipulation Drills

1. AB is parallel to ED . Given $\angle ABC = 130^\circ$ and $\angle CDE = 150^\circ$, find $\angle BCD$.

- A) 60°
 B) 70°
 C) 80°
 D) 100°
 E) 110°



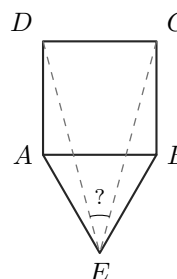
Answer: C) 80°

Method: Draw the line through C parallel to AB . Co-interior angles with $\angle ABC = 130^\circ$ give 50° above it; co-interior angles with $\angle CDE = 150^\circ$ give 30° below it. So $\angle BCD = 50^\circ + 30^\circ = 80^\circ$.

Investigate further: For any zig-zag between parallel lines, the angle at the kink equals the sum of the two outer angles measured on the same side. Rewrite the rule using the reflex angles at B and D and check it still gives 80° .

2. $ABCD$ is a square and ABE is an equilateral triangle drawn outside the square on side AB . What is $\angle DEC$?
 (after Primer 1.4 P11)

- A) 15°
 B) 20°
 C) 45°
 D) 60°
 E) 30°



Answer: E) 30°

Method: $AD = AB = AE$ and $\angle DAE = 90^\circ + 60^\circ = 150^\circ$, so triangle ADE is isosceles with base angles 15° . By symmetry $\angle BEC = 15^\circ$. Hence $\angle DEC = 60^\circ - 15^\circ - 15^\circ = 30^\circ$.

Investigate further: Put E inside the square instead. Show that $\angle DEC$ becomes 150° : the base angles of ADE are now 75° .

3. The triangle with sides 13, 14, 15 has area 84. Its inscribed circle has what radius?

Answer: 4

Method: $r = \frac{A}{s} = \frac{84}{21} = 4$.

Investigate further: Find the circumradius of the same triangle using $R = \frac{abc}{4[A_{BC}]}$, and check that $R \geq 2r$.

4. The five diagonals of a regular pentagon form a five-pointed star whose points are the pentagon's vertices. What is the angle at each point of the star?

- A) 36°
 B) 30°
 C) 45°
 D) 54°
 E) 72°

Answer: A) 36°

Method: The vertices lie on a circle and the five sides are equal chords, so each side subtends the same angle at every other vertex. At each vertex the two diagonals therefore split the 108° into three equal parts, and the point angle is 36° . Check with B10: the five point angles sum to 180° , and $5 \times 36^\circ = 180^\circ$.

Investigate further: For a regular seven-pointed star $\{7/3\}$, joining every third vertex of a heptagon, find the angle at each point.

5. $ABCDEF$ is a regular hexagon. What fraction of its area is triangle ACE ?

- A) $\frac{1}{3}$
- B) $\frac{1}{2}$
- C) $\frac{2}{3}$
- D) $\frac{3}{4}$
- E) $\frac{1}{4}$

Answer: B) $\frac{1}{2}$

Method: Cut the hexagon into six equilateral triangles about its centre. Each corner triangle ABC , CDE , EFA has two sides equal to the hexagon's side and a 120° angle between them, so it has the same area as one of the six equilateral triangles: $\frac{1}{6}$ of the hexagon. So $[ACE] = 1 - 3 \cdot \frac{1}{6} = \frac{1}{2}$.

Investigate further: What fraction of the hexagon is the smaller hexagon formed where the two triangles ACE and BDF overlap?

6. A square, a regular hexagon and one other regular polygon fit together around a point with no gaps or overlaps. How many sides does the other polygon have?

- A) 8
- B) 10
- C) 15
- D) 12
- E) 18

Answer: D) 12

Method: The angles at the point sum to 360° : $90^\circ + 120^\circ + x = 360^\circ$, so $x = 150^\circ$. An interior angle of 150° means an exterior angle of 30° , so 12 sides.

Investigate further: Find all the ways three regular polygons (not necessarily different) can meet at a point with no gaps. There are ten.

7. Each interior angle of a regular polygon is four times each exterior angle. How many sides does it have?

- A) 10
- B) 8
- C) 9
- D) 12
- E) 16

Answer: A) 10

Method: Interior + exterior = 180° and interior = $4 \times$ exterior, so the exterior angle is 36° and there are $\frac{360}{36} = 10$ sides.

Investigate further: For which whole numbers k is there a regular polygon whose interior angle is k times its exterior angle?

8. $ABCDEFGH$ is a regular octagon. Find $\angle ACF$.

Answer: 67.5°

Method: The octagon's vertices lie on a circle, and each side subtends $\frac{360^\circ}{8} = 45^\circ$ at the centre, so 22.5° at any other vertex. The arc AF not containing C runs $A \rightarrow H \rightarrow G \rightarrow F$, three sides, so $\angle ACF = 3 \times 22.5^\circ = 67.5^\circ$.

Investigate further: Without circles: triangle ABC is isosceles with apex 135° , so $\angle BCA = 22.5^\circ$. Finish the angle chase to $\angle ACF$ this way.

9. A regular pentagon and a regular hexagon share a side and lie on opposite sides of it. At each end of the shared side there is a gap between the two shapes. What is the angle of this gap?

- A) 120°
- B) 126°
- C) 128°
- D) 132°
- E) 144°

Answer: D) 132°

Method: At an end of the shared side the angles are the pentagon's 108° , the hexagon's 120° and the gap: $360^\circ - 108^\circ - 120^\circ = 132^\circ$.

Investigate further: Which regular polygon would fill the gap exactly? (None does. Explain why, and find the n -gon that fits the gap left by a square and a regular octagon.)

10. A five-pointed star is drawn with five straight strokes, not necessarily regular. Write down the sum of the angles at its five points.

Answer: 180°

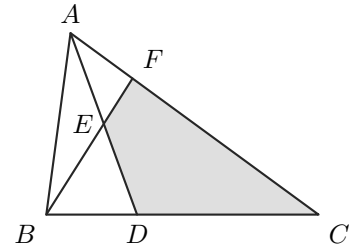
Method: Label the points A, B, C, D, E in order round the star, so the strokes are AC, CE, EB, BD, DA , with point angles a, b, c, d, e . Let BE meet AC at X and AD at Y . Triangle XCE has angles c and e at C and E , so its exterior angle $\angle AXY = c + e$; likewise triangle YBD gives $\angle AYX = b + d$. The angles of triangle AXY then give $a + (c + e) + (b + d) = 180^\circ$.

Investigate further: What is the sum of the point angles of a seven-pointed star drawn by joining every third vertex of a heptagon? (It is 180° for the pentagram, but not in general.)

Section C Substitution & Structure

1. In triangle ABC , the point D lies on BC with $BD : DC = 1 : 2$, and E is the midpoint of AD . The line BE meets AC at F . What fraction of triangle ABC is the shaded quadrilateral $EDCF$?

- A) $\frac{1}{2}$
 B) $\frac{5}{8}$
 C) $\frac{7}{12}$
 D) $\frac{2}{3}$
 E) $\frac{3}{4}$



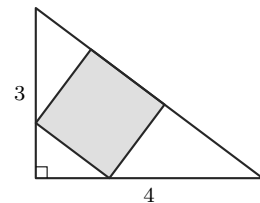
Answer: C) $\frac{7}{12}$

Method: Mass points: $BD : DC = 1 : 2$ puts mass 2 at B and 1 at C , so 3 at D . Since E is the midpoint of AD , A also gets mass 3. On AC this gives $AF : FC = 1 : 3$. Then $[AEF] = \frac{1}{2} \cdot \frac{1}{4} [ADC] = \frac{1}{8} \cdot \frac{2}{3} = \frac{1}{12}$ of $[ABC]$, and $[EDCF] = [ADC] - [AEF] = \frac{2}{3} - \frac{1}{12} = \frac{7}{12}$. Option D is $[ADC]$ with the corner AEF left in.

Investigate further: Redo the question with $BD : DC = 1 : k$ and find $AF : FC$ in terms of k .

2. A right-angled triangle has legs 3 and 4. A square sits inside it with one side along the hypotenuse and its other two vertices on the legs. What is the side length of the square?

- A) $\frac{12}{7}$
 B) $\frac{5}{3}$
 C) $\frac{3}{2}$
 D) $\frac{60}{37}$
 E) $\frac{60}{49}$



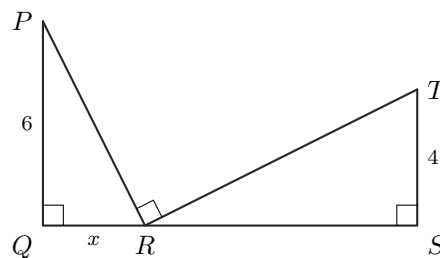
Answer: D) $\frac{60}{37}$

Method: The altitude to the hypotenuse is $h = \frac{3 \cdot 4}{5} = \frac{12}{5}$. The strip of the triangle beyond the square is similar to the whole triangle, with base s and height $h - s$. So $\frac{s}{5} = \frac{h - s}{h}$, giving $s = \frac{5h}{5 + h} = \frac{60}{37}$. Option A, $\frac{12}{7} = \frac{ab}{a+b}$, is the square tucked into the right-angle corner.

Investigate further: Prove that the corner square is always the larger of the two, for every right-angled triangle.

3. PQ and TS are perpendicular to QS , with $PQ = 6$, $TS = 4$ and $QS = 11$. The point R lies on QS with $QR = x$, and $\angle PRT = 90^\circ$. Then (after Primer 1.4 P48)

- A) x has two possible values whose sum is 12
 B) x has two possible values whose difference is 5
 C) x has only one value, and $x \geq 5.5$
 D) x has only one value, and $x < 5.5$
 E) x cannot be determined from the given information



Answer: B) x has two possible values whose difference is 5

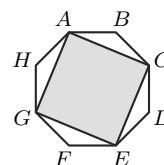
Method: $\angle PRT = 90^\circ$ makes triangles PQR and RST similar, so $\frac{PQ}{QR} = \frac{RS}{ST}$, giving $x(11 - x) = 24$. Then $x = 3$ or $x = 8$, which differ by 5.

Investigate further: For which lengths QS is there exactly one position of R ? Link the answer to the discriminant, and to the circle on diameter PT touching QS .

4. $ABCDEFGH$ is a regular octagon with sides of length

1. What is the area of the shaded square $ACEG$?

- A) $2 + \sqrt{2}$
 B) $1 + \sqrt{2}$
 C) $2 + 2\sqrt{2}$
 D) 4
 E) $3 + 2\sqrt{2}$



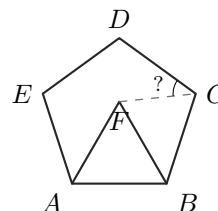
Answer: A) $2 + \sqrt{2}$

Method: Extend four alternate sides to form a square of side $1 + \sqrt{2}$: the octagon is that square with four corner triangles of legs $\frac{1}{\sqrt{2}}$ removed, so its area is $(1 + \sqrt{2})^2 - 4 \cdot \frac{1}{4} = 2 + 2\sqrt{2}$. The square $ACEG$ is the octagon minus four triangles like ABC , each with sides 1, 1 and included angle 135° , area $\frac{1}{2} \sin 135^\circ = \frac{\sqrt{2}}{4}$. So $[ACEG] = 2 + 2\sqrt{2} - \sqrt{2} = 2 + \sqrt{2}$. Option C is the whole octagon; option E is the enclosing square.

Investigate further: What fraction of the octagon is the square $ACEG$? Compare it with the fraction of a regular hexagon taken by triangle ACE (B5).

5. $ABCDE$ is a regular pentagon, and ABF is an equilateral triangle with F inside the pentagon. What is $\angle FCD$?

- A) 36°
 B) 48°
 C) 42°
 D) 54°
 E) 66°



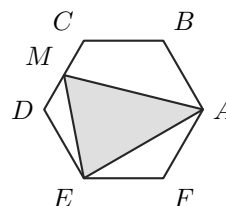
Answer: C) 42°

Method: Each angle of the pentagon is 108° , so $\angle FBC = 108^\circ - 60^\circ = 48^\circ$. Also $BF = BA = BC$, so triangle BFC is isosceles with $\angle BCF = \frac{1}{2}(180^\circ - 48^\circ) = 66^\circ$. Hence $\angle FCD = 108^\circ - 66^\circ = 42^\circ$. Option B is $\angle FBC$; option E is $\angle BCF$.

Investigate further: Find $\angle CFD$, and check that the four angles at F outside the equilateral triangle add up to 300° .

6. $ABCDEF$ is a regular hexagon and M is the midpoint of CD . What fraction of the hexagon is the shaded triangle AME ?

- A) $\frac{1}{3}$
 B) $\frac{1}{2}$
 C) $\frac{1}{4}$
 D) $\frac{3}{8}$
 E) $\frac{5}{12}$



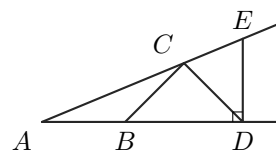
Answer: E) $\frac{5}{12}$

Method: All three triangles ACE , AME , ADE share the base AE . Since M is the midpoint of CD , its distance from the line AE is the average of the distances of C and D , so $[AME] = \frac{1}{2}([ACE] + [ADE])$. Now $[ACE] = \frac{1}{2}$ of the hexagon (B5), and $[ADE] = \frac{1}{3}$: its base AD is twice a side and its height is that of one equilateral triangle, so it is two of the six. Hence $[AME] = \frac{1}{2}(\frac{1}{2} + \frac{1}{3}) = \frac{5}{12}$. Option A is $[ADE]$; option B is $[ACE]$.

Investigate further: As M moves along the side CD , how does $[AME]$ change? Where on the hexagon's boundary should M be to make $[AME]$ as large as possible?

7. The points B and D lie on one arm of an angle at A , and C and E lie on the other arm, with $AB = BC = CD = DE$. The line DE is perpendicular to AD . What is $\angle BAC$?

- A) 15°
 B) 18°
 C) 20°
 D) 22.5°
 E) 30°



Answer: D) 22.5°

Method: Let $\angle BAC = x$. Triangle ABC is isosceles, so $\angle BCA = x$ and the exterior angle $\angle CBD = 2x$. Triangle BCD is isosceles, so $\angle CDB = 2x$. At C the angles along the arm give $\angle DCE = 180^\circ - x - (180^\circ - 4x) = 3x$, and triangle CDE is isosceles, so $\angle CED = 3x$. In triangle ADE : $x + 90^\circ + 3x = 180^\circ$, so $x = 22.5^\circ$.

Investigate further: Keep adding equal segments EF , FG , ... alternately on the two arms. If there are k equal segments and the last one is perpendicular to an arm, what is x ? What is the largest number of equal segments that fit for a given x ?

8. A regular 12-sided polygon is inscribed in a circle of radius 1. What is the total area inside the circle but outside the polygon?

- A) $\pi - 2$
 B) $\pi - 3$
 C) $\pi - \frac{3\sqrt{3}}{2}$

D) $\pi - 2\sqrt{2}$

E) $\frac{\pi}{12}$

Answer: B) $\pi - 3$

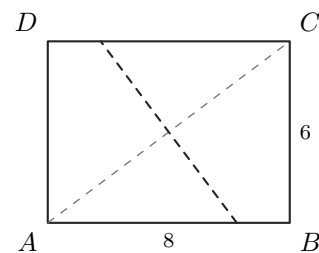
Method: The dodecagon is twelve isosceles triangles with two sides 1 and apex angle 30° , each of area $\frac{1}{2} \sin 30^\circ = \frac{1}{4}$. So its area is 3, and the region outside it is $\pi - 3$. Option A uses the square, option C the hexagon, option D the octagon.

Investigate further: Show that the regular 12-gon inscribed in a circle of radius R has area exactly $3R^2$, and use it to prove $\pi > 3$.

Section D Challenge

1. A rectangular sheet $ABCD$ has $AB = 8$ and $BC = 6$. It is folded along a straight crease so that C lands exactly on A . How long is the crease?

- A) 7
B) $5\sqrt{2}$
C) $\frac{25}{4}$
D) 8
E) $\frac{15}{2}$

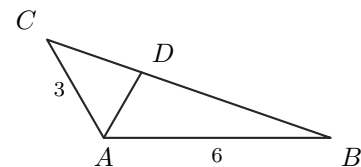
**Answer:** E) $\frac{15}{2}$

Method: The crease is the perpendicular bisector of AC (every point on it is equidistant from A and C), so it passes through the centre M with $AM = 5$. It meets AC at right angles, so its half-length is $AM \tan \angle CAB = 5 \cdot \frac{6}{8} = \frac{15}{4}$. The full crease is $\frac{15}{2}$.

Investigate further: Show that for an $a \times b$ sheet the crease is $\frac{b\sqrt{a^2 + b^2}}{a}$. What shape does the folded sheet make?

2. In triangle ABC , $AB = 6$, $AC = 3$ and $\angle BAC = 120^\circ$. The bisector of angle A meets BC at D . What is the length of AD ?

- A) $2\sqrt{3}$
B) 2
C) 4
D) $\sqrt{7}$
E) 3

**Answer:** B) 2

Method: Split the area along AD : $[ABD] + [ADC] = [ABC]$, so $\frac{1}{2} \cdot 6 \cdot AD \sin 60^\circ + \frac{1}{2} \cdot 3 \cdot AD \sin 60^\circ = \frac{1}{2} \cdot 6 \cdot 3 \sin 120^\circ$. That gives $9AD = 18$, so $AD = 2$. Option C, $4 = \frac{2bc}{b+c}$, forgets the factor $\cos \frac{A}{2}$.

Investigate further: For a 120° angle this says $\frac{1}{AD} = \frac{1}{AB} + \frac{1}{AC}$. Derive the general bisector length $AD = \frac{2bc \cos(A/2)}{b+c}$.

3. What is the side length of the largest square that fits inside a regular hexagon with sides of length 1?

- A) 1
 B) $\frac{2\sqrt{3}}{3}$
 C) $3 - \sqrt{3}$
 D) $\frac{\sqrt{6}}{2}$
 E) $\sqrt{2}$

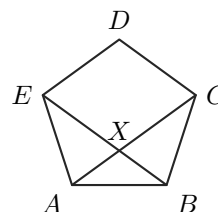
Answer: C) $3 - \sqrt{3}$

Method: Place the hexagon with vertices $(\pm 1, 0)$ and $(\pm \frac{1}{2}, \pm \frac{\sqrt{3}}{2})$, and the square centred at the origin with sides parallel to the axes, so two of its sides are parallel to two sides of the hexagon. A corner $(\frac{s}{2}, \frac{s}{2})$ must lie inside the slanted side $\sqrt{3}x + y = \sqrt{3}$: $\frac{\sqrt{3}+1}{2}s \leq \sqrt{3}$, so $s \leq \frac{2\sqrt{3}}{\sqrt{3}+1} = 3 - \sqrt{3} \approx 1.268$, and then the top side at $y = \frac{s}{2} \approx 0.634 < \frac{\sqrt{3}}{2}$ fits too. Turning the square only makes it worse: at 45° (option D) the best is $\frac{\sqrt{6}}{2} \approx 1.225$.

Investigate further: Show that the square's corners touch four of the hexagon's six sides. Which is larger: the largest square in a regular hexagon of side 1, or the largest equilateral triangle in a square of side 1?

4. $ABCDE$ is a regular pentagon with sides of length 1. The diagonals AC and BE meet at X . What is the length AX ?

- A) $\frac{\sqrt{5}-1}{2}$
 B) $\frac{\sqrt{5}+1}{2}$
 C) $\frac{1}{2}$
 D) $\frac{\sqrt{5}}{2}$
 E) $\frac{3-\sqrt{5}}{2}$

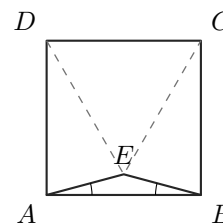


Answer: A) $\frac{\sqrt{5}-1}{2}$

Method: The angles of the pentagon are 108° and each diagonal cuts off an isosceles triangle with base angles 36° . So $\angle XAB = \angle XBA = 36^\circ$, and in triangle BXC , $\angle XBC = 72^\circ$ and $\angle XCB = 36^\circ$, so $\angle BXC = 72^\circ$ and $CX = CB = 1$. Triangles ABC and XAB are similar (angles $36^\circ, 36^\circ, 108^\circ$), so with $d = AC$: $\frac{d}{1} = \frac{1}{AX} = \frac{1}{d-1}$, i.e. $d^2 - d - 1 = 0$, $d = \frac{1+\sqrt{5}}{2}$, and $AX = d - 1 = \frac{\sqrt{5}-1}{2}$. Option B is the whole diagonal.

Investigate further: The five diagonals form a smaller regular pentagon inside. Find its side length, and the ratio of its area to the original pentagon's.

5. $ABCD$ is a square. The point E inside the square satisfies $\angle EAB = \angle EBA = 15^\circ$. Prove that triangle CDE is equilateral.



Answer: Proof: see method

Method: Work backwards from the triangle we want. Construct the equilateral triangle CDF with F inside the square. Then $DF = DC = DA$ and $\angle ADF = 90^\circ - 60^\circ = 30^\circ$, so triangle ADF is isosceles with $\angle DAF = \frac{1}{2}(180^\circ - 30^\circ) = 75^\circ$. Hence $\angle FAB = 90^\circ - 75^\circ = 15^\circ$, and by symmetry $\angle FBA = 15^\circ$. Only one point inside the square makes 15° angles at both A and B (the two rays from A and B cross once), so $F = E$. Therefore CDE is equilateral. ■

Investigate further: Why is “build the answer, then show it coincides” legitimate here? Write down exactly which uniqueness fact the proof relies on. Then try a direct proof by angle chasing alone, and see where it gets stuck.

Top 5 Patterns — Worksheet #4

- Exterior angles sum to 360° .** A regular n -gon has exterior angle $\frac{360^\circ}{n}$ and interior angle $180^\circ - \frac{360^\circ}{n}$; angles meeting at a point sum to 360° . (see A1, A4, B6, B7, B9)
- A regular polygon lives on a circle.** Each side subtends $\frac{360^\circ}{n}$ at the centre and half that at any other vertex, and the polygon splits into n equal isosceles triangles. (see A10, B4, B8, C4, C8)
- Chase angles through isosceles triangles.** Equal sides give equal base angles, and the exterior angle of a triangle is the sum of the two opposite interior angles. (see A2, B1, B2, B10, C5, C7, D5)
- Similar triangles carry ratios.** Lengths scale by k , areas by k^2 ; in a pentagon the diagonal and side satisfy $d^2 = d + 1$. (see A9, C2, C3, D1, D4)
- Area ratios from shared heights.** Triangles with the same height have areas in the ratio of their bases; cut regular polygons into equal triangles and count. (see A6, B5, C1, C6, D2)

Common Traps — Worksheet #4

- Using the interior angle where the exterior one is needed.** $\frac{360}{\text{interior}}$ is not the number of sides; divide 360° by the exterior angle. (see A1, A4, B7)
- Stopping at the first isosceles triangle.** In chains and pentagons the answer is usually two or three triangles further on; options like 48° and 66° in C5 are the intermediate angles. (see B2, C5, C7)
- Scaling areas by k instead of k^2 .** Similar figures with lengths in ratio $2 : 3$ have areas in ratio $4 : 9$. (see A9, C2, C4)
- Forgetting a corner piece.** The shaded region is often the obvious triangle minus (or plus) a small corner: $\frac{2}{3}$ in C1 and $2 + 2\sqrt{2}$ in C4 are the uncorrected answers. (see C1, C4, C6, C8)
- Assuming the diagram is to scale or the obvious orientation is best.** The largest

square in a hexagon is not the tilted one, and a crease is not a diagonal. (see $D1$, $D3$, $D4$)

“Every polygon is a bundle of triangles: count the exterior angles, find the isosceles pairs, and let similar copies carry the ratios.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths